# C29xMCAL_01.02.01 Release Notes


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# Introduction

F29H85x package contains MCAL and CDD modules as per AUTOSAR architecture. These software modules are developed specific for platform hardware to enable the direct access to on-chip MCU peripherals. Thus making the upper layer of AUTOSAR architecture independent of MCU platform hardware. These drivers are developed as per AUTOSAR version 4.3.1. The delivery contains source code of all drivers, EB Tresos plugin to generate configuration data, bare-metal examples of each driver and its user guide.

**F29H85x MCAL v01.02.01 is an early adopter patch release and not functional safety compliant targeted for F29H85x.**

# Device Support

The following devices are supported with this release

- MCU's
  - F29H859TU-Q1

- Hardware Platforms
  - F29H85X-SOM-EVM

# What Is Supported

- MCAL / CDD (AUTOSAR **v4.3.1**):

| **Drivers** | **Modules** |
|---|---|
| Micro controller Drivers | GPT, MCU, WDG |
| Communication Driver | CAN, LIN, SPI |
| I/O Drivers | DIO, PORT |
| Memory Drivers | FLS |
| Complex Device Drivers | ADC, XBAR, SENT, IPC, UART, PWM, ECAP |

- Reference Code:
  - Multi-Core startup code
- Build System:
  - CMake
  - CCS Projectspec

**Note: The minimum version of CCS required is CCSv20.01.01 and compiler version is C29Clang 1.0.0 LTS. Please refer to the Dependencies section for the download links.**

# What Is Not Supported

- Multi-core MCALs as per AUTOSAR 4.3.1.

# New In This Release

- MCAL / CDD :
  - Existing modules with bug fixes -
    - 
    - PORT
    - CAN
    - FLS
    - DIO
    - CAN
    - CDD IPC
    - CDD XBAR
    - CDD ECAP
    - CDD SENT

- CDD UART

# Module Version Number

| MCAL / CDD | Module Version Number |
|---|---|
| MCU | 01.01.00 |
| PORT | 01.01.01 |
| DIO | 01.00.03 |
| GPT | 01.00.02 |
| CAN | 01.02.00 |
| WDG | 01.00.01 |
| LIN | 01.01.00 |
| SPI | 01.01.01 |
| FLS | 01.01.00 |
| CDD ADC | 02.00.00 |
| CDD XBAR | 02.00.01 |
| CDD SENT | 02.00.01 |
| CDD PWM | 01.00.00 |
| CDD ECAP | 01.00.01 |
| CDD IPC | 01.00.01 |
| CDD UART | 01.00.01 |

**Note:**

We follow Autosar's requirement [SRS_BSW_00321] for versionoing of each module-

**5.2.4.10 [SRS_BSW_00321] The version numbers of AUTOSAR Basic Software Modules shall be enumerated according specific rules**

[

| | |
|---|---|
| ***Type:*** | Valid |
| ***Description:*** | The version numbers of AUTOSAR Basic Software Modules shall be enumerated according to the following rules:<br>- Increasing a more significant digit of a version number resets all less significant digits<br>- The PATCH_VERSION is incremented if the module is still upwards and downwards compatible (e.g. bug fixed)<br>- The MINOR_VERSION is incremented if the module is still downwards compatible (e.g. validfunctionality added)<br>- The MAJOR_VERSION is incremented if the module is not compatible any more (e.g. existing API valid)] |
| ***Rationale:*** | Provide unambiguous version identification for each module, provide version cross check as well as basic version retrieval facilities.<br>Compatibility is always visible! |
| ***Use Case:*** | Example: ADC module with version 1.14.2: |

# Fixed In This Release

| Key | Summary | Component/s | Severity |
|---|---|---|---|
| MCAL-30801 | Port configuration is incorrect for the example Gpt_Example_InterruptTimeout | Gpt | S3-Minor |
| MCAL-30735 | READONLY parameters are not auto-calculated via Tresos CLI | CDD IPC, CDD SENT, CDD UART, Dio, Fls, Port | S3-Minor |
| MCAL-30727 | MCAL driver shall provide flexible type definition for OS Counter | Can | S3-Minor |
| MCAL-30721 | Header protection in Memmap.h files is incorrect | Common | S3-Minor |
| MCAL-30700 | Not all the Inputs Xbar were present in the input list of CDD ECAP | CDD ECAP | S3-Minor |
| MCAL-30695 | DioPortId validation code in incorrect in DIO module | Dio | S2-Major |
| MCAL-30632 | Syntax error in Cdd_Uart_Cfg.c file in CDD UART module | CDD UART | S3-Minor |
| MCAL-30631 | PortPinId validation code in incorrect in PORT module | Port | S2-Major |
| MCAL-30608 | When the number of transmitted data bytes is 6, the CAN transmission data is incorrect | Can | S2-Major |
| MCAL-30597 | CanIf_TxConfirmation is always called with Pdu Id zero in INTERRUPT mode | Can | S2-Major |

| | | | |
|---|---|---|---|
| MCAL-30346 | CDD ECAP: Signal measurements are incorrect for signals in ms period range | CDD ECAP | S1-Critical |
| MCAL-30343 | Mcu module defines assembly code using asm instead of __asm | Wdg | S3-Minor |
| MCAL-30342 | Can defines Enums without the <mip> prefix | Can | S3-Minor |
| MCAL-30308 | CDD ECAP: Missing timestamps in buffer | CDD ECAP | S2-Major |
| MCAL-30268 | MainFunction Write schedulable entity and timing event is not generated in Cdd_Uart_BSWMD.arxml | CDD UART | S3-Minor |
| MCAL-30149 | CDD ECAP: First rising edge not detected/not counted in BOTH_EDGE activation condition | CDD ECAP | S3-Minor |

16 issues

# Known Issues And Limitations

There are few limitations and known issues associated with this release

## Limitations

- CDD PWM - Following sub-modules are not supported:
  - Dead-band Submodule
  - Chopper Submodule
  - Trip-Zone Submodule
  - Diode Emulation Submodule
  - MINDB & ICL submodules
  - Event-trigger Submodule
  - Digital Compare Submodule

- CDD ECAP -
  - Cdd Ecap driver doesn't support APWM mode (single channel PWM generation from ECAP)
  - Cdd Ecap driver doesn't support ECAP Synchronization (SYNCIN)
  - Cdd Ecap doesn't support Signal monitoring units
  - Cdd Ecap doesn't operate ECAP in Single-shot capture of up to four event time-stamps

- Test Infrastructure : Following tests have not been performed due to limitations of available test infrastructure -
  - CDD SENT : Following modes in SENT are not tested -
    - Enhanced 16-bit Slow Mode.
    - Nibble Count 1,2,3,4,5.
    - MTP Mode with all 4 sensors connected.
  - CDD IPC : IPC is validated between CPU1 and CPU3. Since, CPU2 is configured in lockstep in our framework; IPC between CPU1 and CPU3 or CPU2 and CPU3 is not validated.
  - Fls : Timeout tests are not performed as FLASH characterization data is not yet available for this device.
  - Spi : Board to board Spi communication validated upto 40MHz. 50MHz have been only test in loopback mode.

## New Known Issues

| Key | Summary | Component/s | Workaround | Severity |
|---|---|---|---|---|

No issues found

## Existing Known Issues

| Key | Summary | Component/s | Workaround | Severity |
|---|---|---|---|---|
| MCAL-30035 | Mixed mode handling is incorrect in Can | Can | The processing mode for transmit/receive message can be configured to either polling or interrupt for Can controller. | S3-Minor |

1 issue

# Dependencies

This release has dependency on the following tools

- CCS v20.01.01.
- C29Clang Compiler 1.0.0 LTS.
- EB Tresos (ACG 8.8.10) : Request access here.

# References

1. F29H85x MCAL User Guide
2. F29H85x Technical Reference Manual and Device Datasheet : F29H859TU-Q1.

# Technical Support

C2000 microcontroller e2e forum